

PROJECT A.E.G.I.S.

Adaptive Execution and Goal-oriented Intelligence System

Competitive Landscape Analysis & Differentiation Report

Existing Industry Players, Their Current Technical Focus, and Concrete Directions Where AEGIS Can
Deliver Original Contribution

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Executive Summary

The AI-powered desktop copilot space has moved extremely quickly since early 2025. What used to be a handful of experimental research prototypes (Google's Project Mariner, OpenAI's Operator) is now, as of mid-2026, a set of generally-available enterprise and consumer products from Microsoft, Google, Apple, Anthropic, OpenAI and Perplexity — several of which now include direct “computer use” capability: an AI agent that can see a screen, move a mouse, click, type and operate applications the way a human would.

This report has two goals. First, it maps exactly what each major company is building today, so the AEGIS proposal is grounded in the current state of the art rather than the 2023–2024 generation of assistants (Siri, Alexa, Cortana) referenced in the original literature review. Second, it identifies twelve concrete, actionable directions where AEGIS can differentiate itself from all of these players — several of which are realistic to implement within a final-year-project timeline, and which map directly onto the research gaps (G1–G5) already identified in the proposal.

The core finding is this: the industry's computer-use agents are overwhelmingly built for enterprise workflow automation (replacing RPA on legacy systems with no API) or for ecosystem-locked consumer convenience (Windows, iOS, Workspace). None of them is built as a personal, transparent, memory-driven, privacy-first academic assistant for an individual student. That is the open space AEGIS can occupy.

1. Existing Companies at a Glance

The table below summarises the major companies currently active in AI-powered desktop copilots, agentic computer-use automation, and context-aware personal assistance — the same problem space targeted by AEGIS.

Company	Product	What It Does	Overlap with AEGIS
Microsoft	Copilot / Copilot Studio (Computer Use)	Embedded assistant across Windows, Microsoft 365 and Power Platform. Summarises documents, drafts emails, and — since May 2026 — operates full desktop/web UIs autonomously via vision-based “computer use” agents, including legacy apps with no API.	Very High
Apple	Apple Intelligence	On-device AI woven into macOS and iOS. Uses personal context (mail, messages, calendar) to power a smarter Siri, writing tools, and notification/email summarisation across first- and third-party apps.	High
Google	Gemini (Workspace, Chrome, Android)	Multimodal assistant embedded in Gmail, Docs, Chrome and Android. Google's dedicated browser-automation agent, Project Mariner, was discontinued in May 2026, with its capabilities folded into Gemini and Chrome directly.	High
Anthropic	Claude Computer Use / Cowork	Gives Claude direct control of a computer: moving the mouse, clicking, typing, and operating a browser or desktop inside a sandboxed environment. Cowork extends this into a multi-step, tool-using work agent for knowledge tasks.	Very High
OpenAI	ChatGPT Agent (formerly Operator) / Atlas	A Computer-Using Agent (CUA) that browses, plans, and executes multi-step web tasks. The standalone Operator product was retired in 2025; its engine now powers ChatGPT's Agent Mode and the dedicated Atlas browser.	High
Perplexity AI	Comet Browser / Assistant	An agentic browser that performs in-page research, summarisation and autonomous multi-step tasks (bookings, email management, form-filling) on top of local browsing context.	Moderate–High

A general observation across all six: every one of them is anchored to a specific commercial ecosystem (Windows, Apple devices, Google Workspace, or a paid chat subscription). None is designed as an independent, ecosystem-agnostic personal copilot that a student can run locally, inspect, and fully control — which is precisely the architectural premise of AEGIS.

2. Detailed Analysis — What Each Company Is Actually Building

2.1 Microsoft — Closest Architectural Overlap

Microsoft is the single closest competitor to the AEGIS concept, because it is the only company pursuing full OS-level context, memory and computer-use automation simultaneously, across both consumer Windows and enterprise Power Platform. Its current work spans:

- Windows Copilot — a system-wide conversational assistant integrated into the OS shell.
- Recall — an on-device, searchable timeline of everything a user has seen on their PC.
- Click to Do — contextual actions triggered directly from on-screen content.
- Copilot Studio Computer Use — reached general availability on 13 May 2026, giving agents a browser, screen and keyboard so they can operate any desktop or web application by reading the UI visually, rather than relying on brittle, hard-coded selectors.
- Windows Agents — local agents, announced at Build 2026, that can manipulate desktop apps via UI automation but run inside a sandbox and remain subject to group policy controls.
- Copilot Tasks — a separate, cloud-hosted agent that executes long-running background tasks (e.g. “schedule weekly team syncs”) on its own dedicated virtual machine.

What Recall does, concretely: It builds a locally indexed, searchable history of on-screen content over time.

Illustrative Example

You viewed a PDF about machine learning yesterday.

Today you type: “the machine learning PDF.”

Recall retrieves it instantly, without you remembering the file name or location.

Microsoft's computer-use agents currently support only native Windows frameworks (WinForms, WPF, UWP, WinUI, Win32) plus browser-based sites with credential entry. Electron apps (VS Code, Slack, Discord), Java-based enterprise software, and Citrix-published apps are explicitly unsupported — a real technical gap AEGIS could target if it chooses a broader automation backend.

2.2 Apple — Personal Context, On-Device Priority

Apple's approach centres on tightly-scoped personal context and on-device processing rather than broad autonomous action-taking. Current focus areas include:

- Personal Context — using a user's own mail, messages and calendar to ground responses.
- Smart Siri — deeper, more conversational query handling.
- Email and Notification Summarisation — condensing high volumes of incoming information.
- Writing Assistant — rewriting, proofreading and tone adjustment system-wide.
- Cross-App AI — surfacing the same assistant capabilities consistently inside first- and third-party apps.

Illustrative Example

You say: “Rewrite this email politely.”

Apple Intelligence rewrites it in place, using the surrounding context it already has access to.

Apple deliberately avoids autonomous multi-step desktop automation of the kind Microsoft, OpenAI and Anthropic pursue — its assistant remains largely reactive and confined to Apple's own app ecosystem, which is a meaningful point of contrast with AEGIS's cross-application ambitions.

2.3 Google — Multimodal, Ecosystem-Wide Assistance

Google's investment centres on the Gemini model family embedded across its own product surface:

- Gemini in Gmail, Docs, Chrome and Android.
- Multimodal understanding — text, image, audio and video in a single model.

- Gemini Enterprise Agent Platform (formerly Vertex AI/Agentspace) for enterprise-grade agent orchestration, with a growing catalogue of partner agents (Box, Workday, Salesforce, ServiceNow).

Notably, Google's dedicated browser-automation research prototype, Project Mariner, was quietly discontinued on 4 May 2026. Google's own stated reasoning was that browser-rendering agents were technically impressive but slow and costly compared to lighter, CLI-style automation approaches now favoured across the industry; Mariner's capabilities were folded into Gemini Agent and Chrome's built-in automatic navigation instead. This is a useful signal for AEGIS: a full browser-rendering approach to automation is expensive to operate at scale, and a leaner, more targeted automation layer may be both more original and more feasible for a student project.

2.4 Anthropic — The Computer-Use Pioneer

Anthropic's Claude Computer Use capability, and its newer Cowork agent product, represent the architecture that overlaps most directly with AEGIS's Action Execution Layer and Task Planning Agent. It can:

- Move the mouse and click on-screen elements.
- Type using the keyboard.
- Fill in forms.
- Operate a full browser session.

Cowork additionally runs agent execution inside isolated sandboxes (Docker containers and virtual machines), and Anthropic layers real-time safety classifiers on top to detect and block potential misuse before an action executes. This sandboxed, classifier-gated approach is conceptually very close to AEGIS's proposed Safety Layer (Module 7) — which strengthens the case that AEGIS's specific contribution should be less about the raw computer-use mechanism (already well-solved by Anthropic and Microsoft) and more about what AEGIS does with persistent context and memory around that mechanism.

2.5 OpenAI — Agentic LLMs and Tool Calling

OpenAI's strategy is centred on general-purpose agentic capability inside its LLMs rather than a dedicated desktop product:

- Large language models as the reasoning core.
- Agentic AI — planning and executing multi-step tasks, not just answering questions.
- Tool calling — invoking external functions, APIs and browsers as needed.
- Memory — limited, account-level personalisation across chats.
- Automation — the Computer-Using Agent (CUA) that originally powered the standalone Operator product.

Operator itself was retired as a standalone product in 2025 after struggling with complex JavaScript-heavy sites, CAPTCHAs and session management; its underlying CUA technology now powers ChatGPT's Agent Mode and the dedicated Atlas browser, which reportedly scores around 87% on the WebVoyager web-automation benchmark — one of the strongest published results in the category.

2.6 Perplexity AI — Local, Search-Centric Workflow Agent

Perplexity's Comet browser and assistant focus on combining strong retrieval/search with light autonomous action: in-page research and summarisation, plus multi-step tasks such as bookings, email handling and form-filling, run directly on top of a user's local browsing context. Comet has also begun rolling out to enterprise customers with silent MDM deployment across macOS and Windows — signalling the same shift toward managed, organisation-scale automation seen at Microsoft and Google, rather than a genuinely personal, individually-owned assistant.

3. Mapping the Landscape onto AEGIS's Research Gaps

The AEGIS proposal already identifies five research gaps (G1–G5) in Section 4. Reviewing the current industry landscape confirms that all five remain genuinely open, even against the most advanced 2026 products.

Gap	Research Gap	Why Existing Industry Players Do Not Close It
G1	Holistic Desktop Context Integration	Every major player integrates context within its own ecosystem only (Microsoft → Windows/365, Google → Workspace/Android, Apple → Apple apps). None fuses file system + open processes + browser + clipboard + interaction history into one cross-application retrieval layer that a user fully owns.
G2	Persistent Cross-Session Memory	Enterprise agents (Copilot Studio, Cowork Managed Agents) are session- or organisation-scoped. Consumer assistants personalise loosely via account history. No product offers a transparent, user-controlled episodic + semantic memory store that a student or individual can inspect and edit.
G3	Empirical Cognitive-Load Evaluation	Vendors publish task-benchmark scores (WebVoyager, ScreenSpot) but not controlled, psychometrically validated studies of user cognitive load (e.g. NASA-TLX) for desktop copilots. This remains an open empirical gap AEGIS is positioned to fill.
G4	Human-in-the-Loop Safety for OS-Level Actions	Enterprise tools (Copilot Studio, Managed Agents) have mature governance, but it is built for IT administrators, not individual end-users. A lightweight, individual-friendly risk-tiered approval UX is largely unaddressed.
G5	Proactive vs. Reactive Assistance	Current products are still largely reactive (respond to a prompt) or narrowly proactive within one app (e.g. Recall's search-back). Systematic study of when proactive, cross-app suggestions help vs. disrupt workflow is still an open research question.

4. Twelve Directions Where AEGIS Can Differentiate

The following twelve ideas extend the AEGIS architecture in ways that are not currently offered — at least not together, and not for an individual, privacy-conscious user — by any of the companies reviewed above. Each includes what it is, why it matters relative to the competitive landscape, and a concrete illustration.

4.1 Intelligent Workflow Learning

What it is: Existing assistants execute commands you give them. AEGIS observes a user's repeated routine over time and learns the underlying workflow without being explicitly taught.

Why it matters: None of the six companies reviewed offer this for cross-application, individually-owned routines — Recall lets you search the past, but it does not learn and offer to automate a recurring sequence of actions across unrelated apps.

Illustrative Example

Every day you open: FAST Flex → Gmail → VS Code → GitHub.

After two weeks, AEGIS says: “You use these four applications every morning. Should I launch them automatically?”

4.2 Workflow Suggestions

What it is: Rather than only executing a command, AEGIS predicts and proposes the logical next step in a task.

Why it matters: Reactive command execution is the industry norm (per Research Gap G5); systematically offering the next likely action, grounded in observed context, is still largely unexplored outside single-app copilots like GitHub Copilot's code suggestions.

Illustrative Example

You open GitHub, then VS Code, then edit the README.

AEGIS asks: “Would you like to commit these changes as well?”

4.3 Productivity Analysis Dashboard

What it is: A weekly, automatically generated report on how time was actually spent across applications, with qualitative insights, not just raw numbers.

Why it matters: This directly operationalises RQ3 (cognitive load / workload) with real usage telemetry, going beyond the post-hoc NASA-TLX survey into continuous, passive measurement — something none of the reviewed commercial products expose to the end-user themselves.

Illustrative Example

This week: Chrome – 5 hrs, VS Code – 16 hrs, Social Media – 7 hrs, Deep Work – 12 hrs, Interruptions – 54, Focus Score – 82%.

Insight: “You were notably more productive on Wednesday.” / “You switch applications roughly every 8 minutes.”

4.4 Explain Before Acting

What it is: Before executing a multi-step action, AEGIS states its intended plan in plain language and waits for explicit approval — not just for high-risk actions, but as a default interaction pattern.

Why it matters: This goes beyond the risk-tiered approval systems used by Microsoft (Copilot Studio governance) and OpenAI (Operator's Watch/Takeover modes), which are largely designed for IT administrators. AEGIS can make this transparency legible and comfortable for an individual, non-technical end-user by default.

Illustrative Example

AEGIS: “I will perform these steps: (1) open Gmail, (2) attach Resume.pdf, (3) set subject to ‘Tesla Internship’, (4) send the email. Approve?”

4.5 Smart File Organization

What it is: AEGIS notices duplicate, redundant or messily-named files and proactively offers to clean and organise them.

Why it matters: File-system hygiene is a common real pain point that general-purpose assistants largely ignore since they are not deeply integrated with local storage the way AEGIS's Context Retrieval Module is designed to be.

Illustrative Example

Downloads folder contains: Resume.pdf, Resume(2).pdf, resume_new.pdf, resume_final.pdf, resume_latest.pdf.

AEGIS: “These look like duplicate versions. Should I move them into an ‘Internship Documents’ folder?”

4.6 Personal Knowledge Base

What it is: All of a user's PDFs, notes, assignments and code projects are indexed into a single, semantically searchable personal corpus.

Why it matters: This is a natural extension of the Memory Layer's planned vector database, and directly demonstrates the value of Context Retrieval Module semantic search (Module 3) in a way evaluators can experience firsthand.

Illustrative Example

“Show me the regression assignment.”

“Find the OOP lab that covered inheritance.”

AEGIS retrieves the right file instantly.

4.7 Student Mode

What it is: A dedicated mode that shifts AEGIS's behaviour to support studying and exam preparation specifically, rather than general productivity.

Why it matters: Every company reviewed builds for a general consumer or enterprise employee. None targets a student's academic workflow specifically — this is a genuinely open niche and gives the dissertation a sharply defined, defensible contribution.

Illustrative Example

Exam Mode triggers: detect syllabus → open relevant PDFs → summarise notes → suggest past papers → start a timer → block phone/desktop distractions.

4.8 Multi-Agent System

What it is: Formalising the Task Planning Agent into several specialised, cooperating sub-agents rather than one monolithic planner.

Why it matters: This mirrors an active trend in the wider agentic-AI research literature and gives AEGIS a cleaner story for modularity and extensibility.

Illustrative Example

Planning Agent → Search Agent → Execution Agent → Safety Agent → Memory Agent, each with a narrow,

well-defined responsibility.

4.9 😊 Behavioural Stress Signals (No Biometric Sensing)

What it is: AEGIS infers likely frustration purely from interaction patterns — rapid Alt+Tab, repeated Undo/Delete, frequent window closing — without any camera or biometric input.

Why it matters: This is a genuinely novel, low-cost proactive-assistance trigger (directly relevant to RQ4/G5) that no reviewed competitor exposes as a first-class feature.

Illustrative Example

After repeated Alt+Tab, Delete and Undo actions in a short window, AEGIS asks: “It looks like something isn’t working smoothly. Would you like me to switch to Focus Mode?”

4.10 📈 Self-Improvement Loop

What it is: AEGIS evaluates the outcome of each executed plan and uses failures to refine future planning for similar tasks.

Why it matters: Adds a feedback loop the current architecture does not yet formally include, and gives the Task Planning Agent (Module 4) a concrete mechanism for the “revise plans in response to execution failures” behaviour already mentioned in the proposal.

Illustrative Example

Task → Plan → Execute → Success? → Error? → Feed outcome back into planning for next time.

4.11 🛡️ Privacy Score

What it is: Every time AEGIS accesses a file or data source, it surfaces a clear Low / Medium / High privacy-risk rating with a plain-language reason.

Why it matters: This makes the ethical commitments already stated in Section 14.2 of the proposal (data minimisation, transparency) tangible and visible to the user in real time — something none of the reviewed products expose at this level of granularity to an individual end-user.

Illustrative Example

AEGIS reads a personal finance spreadsheet and flags: “Privacy Risk: High — this file contains financial data and will not be sent to any external service.”

4.12 🌐 Offline / Local AI

What it is: Core functions — note search, file search, and basic desktop automation — work fully offline using a locally hosted model, without dependence on a cloud LLM.

Why it matters: Microsoft, Google, OpenAI and Perplexity are all fundamentally cloud-dependent for their most capable agents. A credible local-inference mode is a genuine point of difference and directly supports the proposal's stated “Local Processing Priority” ethics commitment.

Illustrative Example

A student with no internet access can still search notes, find files, and trigger simple automations, with all inference happening on-device.

5. Conclusion

Microsoft, Apple, Google, Anthropic, OpenAI and Perplexity are all actively building AI-powered desktop and browser agents, and — as of mid-2026 — several of them (particularly Microsoft's Copilot Studio Computer Use and Anthropic's Claude Computer Use / Cowork) are architecturally very close to AEGIS at the mechanical level of “can an AI operate a computer.” That specific capability can no longer be presented as a novel contribution on its own.

However, the review confirms that all six players are built either for enterprise workflow automation at organisational scale, or for general consumer convenience inside a single ecosystem. None combines persistent, user-owned cross-session memory; transparent, individual-friendly human-in-the-loop safety; and a specifically academic, student-oriented use case, evaluated with rigorous cognitive-load methodology. That combination — not the computer-use mechanism itself — is where AEGIS's genuine, defensible research contribution lies, and it maps cleanly onto the five research gaps (G1–G5) already articulated in the proposal.